Sample data used

Program 1 input: 10, 20, 30 (answered n, n, y for end of list)

Program 21 input: k = 3

Programs 8 to 12: file1.txt contains

```
Python is easy to learn
Roll number 11 passed
Learn python programming
Binary 1011 sample
Python lab record
```

Program 1 - Get the Largest Number from a List

```
Enter an integer to the list:
10
Is it end of list?(y/n)
n
Enter an integer to the list:
20
Is it end of list?(y/n)
n
Enter an integer to the list:
30
Is it end of list?(y/n)
y
The List = [10, 20, 30]
Maximum Value in this list: 30
```

Program 2 - Remove Duplicates from a List

```
Duplicated List: [20, 20, 10, 30, 30]
Deduplicated List: [20, 10, 30]
```

Program 3 - Conversion of Tuple to Dictionary

```
Tuple= (('Rajesh', 75), ('Vasudevan', 80), ('Thomas', 85))
Dictionary= {'Rajesh': 75, 'Vasudevan': 80, 'Thomas': 85}
```

Program 4 - Merging Dictionaries

```
Dictionary 1= {'a': 1, 'b': 2}
Dictionary 2= {'c': 3, 'd': 4}
{'a': 1, 'b': 2, 'c': 3, 'd': 4}
```

Program 5 - Checking Common Member in Lists

```
False
```

Program 6 - Find Earlier Date

```
Earlier date is 2025-01-26
```

Program 7 - Subtract 5 Days from Current Date

```
2026-10-10
```

Program 8 - File Copy Operation

Contents of file2.txt after running:

```
Python is easy to learn
Roll number 11 passed
Learn python programming
Binary 1011 sample
Python lab record
```

Program 9 - Capitalise Each Word in a File

Contents of file1.txt after running:

```
PYTHON IS EASY TO LEARN
ROLL NUMBER 11 PASSED
LEARN PYTHON PROGRAMMING
BINARY 1011 SAMPLE
PYTHON LAB RECORD
```

Program 10 - Search and Replace in a File

Contents of file1.txt after running:

```
C is easy to learn
Roll number 11 passed
Learn python programming
Binary 1011 sample
C lab record
```

Program 11 - Count Number of Lines in a File

```
Number of Lines: 5
```

Program 12 - Retrieve Lines Having Two Consecutive 1's

```
['Roll number 11 passed', 'Binary 1011 sample']
Lines containing two consecutive '1's:
Roll number 11 passed
Binary 1011 sample
```

Program 13 - Matrix Operations Using Vectorisation

```
Matrix 1=
 [[1 2 3]
 [1 1 1]
 [1 1 1]]
Matrix 2=
 [[1 0 0]
 [0 1 0]
 [0 0 1]]
Matrix 1 + Matrix 2=
 [[2 2 3]
 [1 2 1]
 [1 1 2]]
Matrix 1 - Matrix 2=
 [[0 2 3]
 [1 0 1]
 [1 1 0]]
Matrix 1 * Matrix 2=
 [[1 2 3]
 [1 1 1]
 [1 1 1]]
2 * Matrix 1=
 [[2 4 6]
 [2 2 2]
 [2 2 2]]
Transpose of Matrix 1=
 [[1 1 1]
 [2 1 1]
 [3 1 1]]
```

Program 14 - Creation of Matrices for 2D Geometric Transformations

```
[[1 0 1]
 [0 1 1]
 [0 0 1]]
[[ 0.8660254 -0.5        0.       ]
 [ 0.5        0.8660254  0.       ]
 [ 0.         0.         1.       ]]
[[2 0 0]
 [0 2 0]
 [0 0 1]]
```

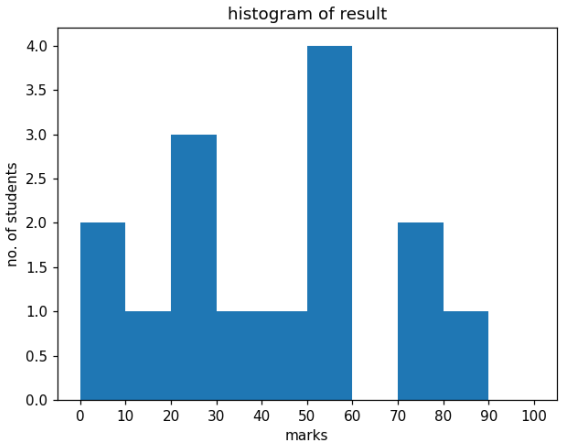
Program 15 - Creation of Matrices for 3D Geometric Transformations

```
[[1 0 0 1]
 [0 1 0 1]
 [0 0 1 1]
 [0 0 0 1]]
[[ 1.         0.         0.         0.       ]
 [ 0.         0.8660254 -0.5        0.       ]
 [ 0.         0.5        0.8660254  0.       ]
 [ 0.         0.         0.         1.       ]]
[[ 0.8660254  0.         0.5        0.       ]
 [ 0.         1.         0.         0.       ]
 [-0.5        0.         0.8660254  0.       ]
 [ 0.         0.         0.         1.       ]]
[[ 0.8660254 -0.5        0.         0.       ]
 [ 0.5        0.8660254  0.         0.       ]
 [ 0.         0.         1.         0.       ]
 [ 0.         0.         0.         1.       ]]
[[2 0 0 0]
 [0 2 0 0]
 [0 0 2 0]
 [0 0 0 1]]
```

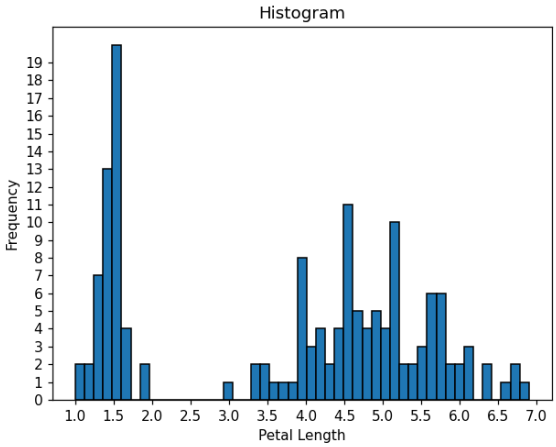
Program 16 - Singular Value Decomposition of a Matrix

```
[[1 2 3]
 [4 5 6]
 [7 8 9]]
[[-0.21483724  0.88723069  0.40824829]
 [-0.52058739  0.24964395 -0.81649658]
 [-0.82633754 -0.38794278  0.40824829]]
[1.68481034e+01 1.06836951e+00 3.33475287e-16]
[[-0.47967118 -0.57236779 -0.66506441]
 [-0.77669099 -0.07568647  0.62531805]
 [-0.40824829  0.81649658 -0.40824829]]
[[1.68481034e+01 0.00000000e+00 0.00000000e+00]
 [0.00000000e+00 1.06836951e+00 0.00000000e+00]
 [0.00000000e+00 0.00000000e+00 3.33475287e-16]]
[[1. 2. 3.]
 [4. 5. 6.]
 [7. 8. 9.]]
```

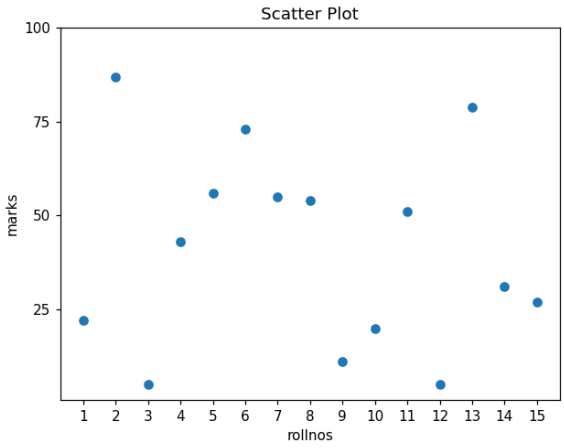
Program 17 - Plot a Histogram



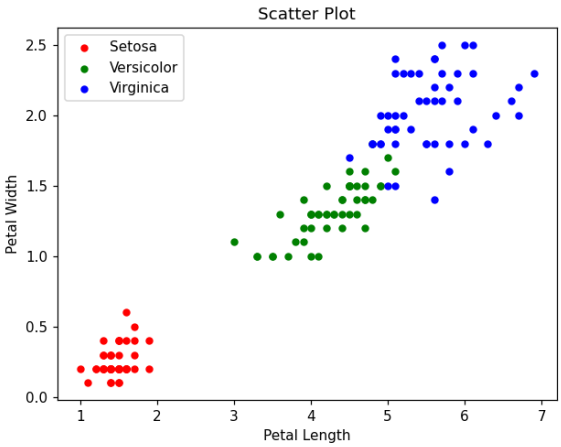
Program 18 - Plot a Histogram Using Iris Data Set



Program 19 - Plot a Scatterplot



Program 20 - Plot a Scatter plot Using Iris Data Set



Program 21 - Predict the Food Type Using k-NN algorithm

```
Enter the value of k:3
Predicted Food Type of Tomato:
[1] fruit
Levels: fruit protein vegetable
```

Program 22 - Classify a Student Using k-NN algorithm

```
Predicted Classification of Student:
[1] Outstanding
Levels: Good Outstanding
```

Program 23 - Prediction of Cancer Using k-NN algorithm

```
  [1] B B B B B B B B B B M B B B B B B B M B B B B M B B B B B M M B M B M B B
 [38] B B B M B B M B B B M M B B B M B B B B B B B B B B B M B M M B B B B B B
 [75] B B B B B B B B B B B B B B B B B B M M M M M M B
Levels: B M
   Cell Contents
|-------------------------|
|                       N |
|           N / Row Total |
|           N / Col Total |
|         N / Table Total |
|-------------------------|


Total Observations in Table:  100


                 | wbcd_test_pred
wbcd_test_labels |         B |         M | Row Total |
-----------------|-----------|-----------|-----------|
               B |        77 |         0 |        77 |
                 |     1.000 |     0.000 |     0.770 |
                 |     0.975 |     0.000 |           |
                 |     0.770 |     0.000 |           |
-----------------|-----------|-----------|-----------|
               M |         2 |        21 |        23 |
                 |     0.087 |     0.913 |     0.230 |
                 |     0.025 |     1.000 |           |
                 |     0.020 |     0.210 |           |
-----------------|-----------|-----------|-----------|
    Column Total |        79 |        21 |       100 |
                 |     0.790 |     0.210 |           |
```

```
-----------------|-----------|-----------|-----------|
```

Program 24 - Prediction of Flu Using Naive Bayes Algorithm

```
Naive Bayes Classifier for Discrete Predictors

Call:
naiveBayes.default(x = symptoms_train, y = symptoms$HasFlu)

A-priori probabilities:
symptoms$HasFlu
    N     Y
0.375 0.625

Conditional probabilities:
               Chills
symptoms$HasFlu         N         Y
              N 0.6666667 0.3333333
              Y 0.4000000 0.6000000

               RunningNose
symptoms$HasFlu         N         Y
              N 0.6666667 0.3333333
              Y 0.2000000 0.8000000

               Headache
symptoms$HasFlu      mild        no    strong
              N 0.3333333 0.3333333 0.3333333
              Y 0.4000000 0.2000000 0.4000000

               Fever
symptoms$HasFlu         N         Y
              N 0.6666667 0.3333333
              Y 0.2000000 0.8000000

Prediction of Flu:
[1] Y
Levels: N Y
```

Program 25 - Prediction of Stealing Using Naive Bayes Algorithm

```
Naive Bayes Classifier for Discrete Predictors

Call:
naiveBayes.default(x = cars_train, y = cars_train_labels)

A-priori probabilities:
cars_train_labels
 no yes
0.5 0.5

Conditional probabilities:
                 Colour
cars_train_labels red yellow
              no  0.4    0.6
              yes 0.6    0.4

                 Type
cars_train_labels SUV sports
              no  0.6    0.4
              yes 0.2    0.8

                 Origin
```

```
cars_train_labels domestic imported
              no       0.6      0.4
              yes      0.4      0.6

Prediction of Stolen Status:
[1] no
Levels: no yes
```

Program 26 - Prediction of Iris Species Using Naive Bayes Algorithm

```
Naive Bayes Classifier for Discrete Predictors

Call:
naiveBayes.default(x = iris_train, y = iris_train_labels)

A-priori probabilities:
iris_train_labels
    Setosa Versicolor  Virginica
 0.3333333  0.3333333  0.3333333

Conditional probabilities:
                 sepal.length
iris_train_labels     [,1]      [,2]
       Setosa     5.025714 0.3266072
       Versicolor 5.894286 0.5455396
       Virginica  6.625714 0.5907836

                 sepal.width
iris_train_labels     [,1]      [,2]
       Setosa     3.445714 0.3567359
       Versicolor 2.782857 0.3468223
       Virginica  2.985714 0.2658426

                 petal.length
iris_train_labels     [,1]      [,2]
       Setosa     1.471429 0.1808012
       Versicolor 4.191429 0.4859021
       Virginica  5.608571 0.5083835

                 petal.width
iris_train_labels      [,1]       [,2]
       Setosa     0.2285714 0.08934872
       Versicolor 1.3228571 0.20448747
       Virginica  2.0485714 0.28218833

 [1] Setosa     Setosa     Setosa     Setosa     Setosa     Setosa
 [7] Setosa     Setosa     Setosa     Setosa     Setosa     Setosa
[13] Setosa     Setosa     Setosa     Versicolor Versicolor Versicolor
[19] Versicolor Versicolor Versicolor Virginica  Versicolor Versicolor
[25] Versicolor Versicolor Versicolor Versicolor Versicolor Versicolor
[31] Versicolor Virginica  Virginica  Virginica  Virginica  Virginica
[37] Virginica  Versicolor Virginica  Virginica  Virginica  Virginica
[43] Virginica  Virginica  Virginica
Levels: Setosa Versicolor Virginica


   Cell Contents
|-------------------------|
|                       N |
|           N / Row Total |
|           N / Col Total |
|         N / Table Total |
```

```
|-------------------------|


Total Observations in Table:  45


                 | iris_test_pred
iris_test_labels |     Setosa | Versicolor |  Virginica |  Row Total |
-----------------|------------|------------|------------|------------|
          Setosa |         15 |          0 |          0 |         15 |
                 |      1.000 |      0.000 |      0.000 |      0.333 |
                 |      1.000 |      0.000 |      0.000 |            |
                 |      0.333 |      0.000 |      0.000 |            |
-----------------|------------|------------|------------|------------|
      Versicolor |          0 |         14 |          1 |         15 |
                 |      0.000 |      0.933 |      0.067 |      0.333 |
                 |      0.000 |      0.875 |      0.071 |            |
                 |      0.000 |      0.311 |      0.022 |            |
-----------------|------------|------------|------------|------------|
       Virginica |          0 |          2 |         13 |         15 |
                 |      0.000 |      0.133 |      0.867 |      0.333 |
                 |      0.000 |      0.125 |      0.929 |            |
                 |      0.000 |      0.044 |      0.289 |            |
-----------------|------------|------------|------------|------------|
    Column Total |         15 |         16 |         14 |         45 |
                 |      0.333 |      0.356 |      0.311 |            |
-----------------|------------|------------|------------|------------|
```

Program 27 - Prediction of Cheating Using C5.0 Decision Tree Algorithm

```
Call:
C5.0.default(x = person_train, y = as.factor(person_train_labels))

Classification Tree
Number of samples: 8
Number of predictors: 4
Tree size: 1

Non-standard options: attempt to group attributes


Call:
C5.0.default(x = person_train, y = as.factor(person_train_labels))


C5.0 [Release 2.07 GPL Edition]  ■Fri Oct 16 07:15:38 2026
-------------------------------

Class specified by attribute `outcome'

Read 8 cases (5 attributes) from undefined.data

Decision tree:
 No (8/2)


Evaluation on training data (8 cases):

■    Decision Tree
■  ----------------
■  Size      Errors
■     1    2(25.0%)   <<
```

```
■   (a)   (b)    <-classified as
■  ----  ----
■     6          (a): class No
■     2          (b): class Yes


Time: 0.0 secs

[1] No No
Levels: No Yes


   Cell Contents
|-------------------------|
|                       N |
|         N / Table Total |
|-------------------------|


Total Observations in Table:  2


                     | person_pred
person_test_labels |        No | Row Total |
-------------------|-----------|-----------|
                No |         1 |         1 |
                   |     0.500 |           |
-------------------|-----------|-----------|
               Yes |         1 |         1 |
                   |     0.500 |           |
-------------------|-----------|-----------|
      Column Total |         2 |         2 |
-------------------|-----------|-----------|
```

Program 28 - Prediction of Cricket Play Using C5.0 Decision Tree Algorithm

```
Call:
C5.0.default(x = cricket_train, y = as.factor(cricket_train_labels))

Classification Tree
Number of samples: 9
Number of predictors: 4
Tree size: 2

Non-standard options: attempt to group attributes


Call:
C5.0.default(x = cricket_train, y = as.factor(cricket_train_labels))


C5.0 [Release 2.07 GPL Edition]  ■Fri Oct 16 07:18:36 2026
-------------------------------

Class specified by attribute `outcome'

Read 9 cases (5 attributes) from undefined.data

Decision tree:

Outlook = Rainy: No (4/1)
Outlook in {Overcast,Sunny}: Yes (5/1)
```

Evaluation on training data (9 cases):

```
■    Decision Tree
■  ----------------
■  Size      Errors

■     2    2(22.2%)   <<


■   (a)   (b)    <-classified as
■  ----  ----
■     3     1    (a): class No
■     1     4    (b): class Yes


■Attribute usage:

■100.00%■Outlook


Time: 0.0 secs

[1] Yes No  Yes Yes Yes
Levels: No Yes


   Cell Contents
|-------------------------|
|                       N |
|           N / Row Total |
|           N / Col Total |
|         N / Table Total |
|-------------------------|


Total Observations in Table:  5


                     | cricket_pred
cricket_test_labels |        No |       Yes | Row Total |
--------------------|-----------|-----------|-----------|
                 No |         0 |         1 |         1 |
                    |     0.000 |     1.000 |     0.200 |
                    |     0.000 |     0.250 |           |
                    |     0.000 |     0.200 |           |
--------------------|-----------|-----------|-----------|
                Yes |         1 |         3 |         4 |
                    |     0.250 |     0.750 |     0.800 |
                    |     1.000 |     0.750 |           |
                    |     0.200 |     0.600 |           |
--------------------|-----------|-----------|-----------|
       Column Total |         1 |         4 |         5 |
                    |     0.200 |     0.800 |           |
--------------------|-----------|-----------|-----------|
```

Program 29 - Identification of Risky Bank Loans Using C5.0 Decision Tree Algorithm

```
Call:
C5.0.default(x = credit_train, y = as.factor(credit_train_labels))

Classification Tree
Number of samples: 900
Number of predictors: 16
Tree size: 63

Non-standard options: attempt to group attributes


Call:
C5.0.default(x = credit_train, y = as.factor(credit_train_labels))


C5.0 [Release 2.07 GPL Edition]  ■Fri Oct 16 07:18:37 2026
-------------------------------

Class specified by attribute `outcome'

Read 900 cases (17 attributes) from undefined.data

Decision tree:

checking_balance in {unknown,> 200 DM}: no (414/53)
checking_balance in {< 0 DM,1 - 200 DM}:
:...months_loan_duration <= 11:
    :...credit_history in {critical,good,poor,perfect}: no (71/11)
    :   credit_history = very good: yes (6/1)
    months_loan_duration > 11:
    :...savings_balance in {unknown,500 - 1000 DM,> 1000 DM}:
        :...checking_balance = 1 - 200 DM: no (52/9)
        :   checking_balance = < 0 DM:
        :   :...savings_balance in {500 - 1000 DM,> 1000 DM}: no (8/1)
        :       savings_balance = unknown:
        :       :...credit_history in {critical,poor,perfect}: no (3)
        :           credit_history = very good: yes (2)
        :           credit_history = good:
        :           :...phone = no: yes (10/1)
        :               phone = yes:
        :               :...job in {skilled,unskilled,unemployed}: no (6/1)
        :                   job = management: yes (2)
        savings_balance in {< 100 DM,100 - 500 DM}:
        :...months_loan_duration > 42: yes (35/5)
            months_loan_duration <= 42:
            :...percent_of_income <= 3:
                :...employment_duration = unemployed:
                :   :...dependents <= 1: no (11/3)
                :   :   dependents > 1: yes (2)
                :   employment_duration = > 7 years:
                :   :...job in {unskilled,unemployed}: no (3)
                :   :   job = management:
                :   :   :...savings_balance = < 100 DM: no (7/2)
                :   :   :   savings_balance = 100 - 500 DM: yes (1)
                :   :   job = skilled:
                :   :   :...savings_balance = 100 - 500 DM: no (1)
                :   :       savings_balance = < 100 DM:
                :   :       :...other_credit in {bank,store}: yes (4)
                :   :           other_credit = none: [S1]
                :   employment_duration = 1 - 4 years:
```

```
:   :...amount > 7721: yes (5)
:   :   amount <= 7721:
:   :   :...housing = own: no (34/8)
:   :       housing = other: yes (1)
:   :       housing = rent:
:   :       :...years_at_residence <= 3: no (5)
:   :           years_at_residence > 3:
:   :           :...credit_history in {critical,good,poor,
:   :               :                  perfect}: yes (7/1)
:   :               credit_history = very good: no (1)
:   employment_duration = < 1 year:
:   :...other_credit = bank: no (3)
:   :   other_credit = store: yes (1)
:   :   other_credit = none:
:   :   :...job in {unskilled,unemployed}: yes (8/1)
:   :       job = management: no (2/1)
:   :       job = skilled:
:   :       :...age > 28: no (5)
:   :           age <= 28:
:   :           :...amount <= 3518: yes (7)
:   :               amount > 3518: no (3)
:   employment_duration = 4 - 7 years:
:   :...savings_balance = 100 - 500 DM: no (8)
:       savings_balance = < 100 DM:
:       :...housing in {other,rent}: no (6)
:           housing = own:
:           :...purpose in {renovations,car0}: yes (0)
:               purpose in {education,business}: no (4)
:               purpose in {furniture/appliances,car}:
:               :...checking_balance = < 0 DM: yes (5)
:                   checking_balance = 1 - 200 DM:
:                   :...months_loan_duration > 18: yes (2)
:                       months_loan_duration <= 18:
:                       :...other_credit in {none,
:                           :               store}: no (3)
:                           other_credit = bank: yes (1)
percent_of_income > 3:
:...credit_history in {perfect,very good}: yes (16/1)
    credit_history in {critical,good,poor}:
    :...purpose = education: yes (6)
        purpose in {business,car0}: no (11/3)
        purpose = renovations:
        :...years_at_residence <= 3: yes (2)
        :   years_at_residence > 3: no (2)
        purpose = car:
        :...savings_balance = 100 - 500 DM:
        :   :...other_credit in {none,store}: no (4)
        :   :   other_credit = bank: yes (1)
        :   savings_balance = < 100 DM:
        :   :...credit_history = poor: yes (2)
        :       credit_history = critical:
        :       :...months_loan_duration > 27: yes (4)
        :       :   months_loan_duration <= 27:
        :       :   :...amount <= 1301: yes (4/1)
        :       :       amount > 1301: no (8/1)
        :       credit_history = good:
        :       :...amount > 1204: yes (12)
        :           amount <= 1204:
        :           :...amount <= 1028: yes (2)
        :               amount > 1028: no (2)
        purpose = furniture/appliances:
        :...savings_balance = 100 - 500 DM: yes (8)
```

```
                            savings_balance = < 100 DM:
                            :...checking_balance = 1 - 200 DM:
                                :...age <= 32: no (12)
                                :   age > 32:
                                :   :...years_at_residence <= 3: yes (4)
                                :       years_at_residence > 3: no (3)
                                checking_balance = < 0 DM:
                                :...years_at_residence <= 1: no (7/1)
                                    years_at_residence > 1:
                                    :...months_loan_duration > 16: yes (22/4)
                                        months_loan_duration <= 16:
                                        :...other_credit in {none,
                                            :               bank}: no (7/1)
                                            other_credit = store: yes (1)

SubTree [S1]

purpose in {furniture/appliances,education,renovations,car0}: no (6/1)
purpose in {car,business}: yes (5/1)


Evaluation on training data (900 cases):

■    Decision Tree
■  ----------------
■  Size      Errors

■    62  112(12.4%)   <<


■   (a)   (b)    <-classified as
■  ----  ----
■   616    16    (a): class no
■    96   172    (b): class yes


■Attribute usage:

■100.00%■checking_balance
■ 54.00%■months_loan_duration
■ 45.44%■savings_balance
■ 32.33%■percent_of_income
■ 27.56%■credit_history
■ 16.78%■employment_duration
■ 16.67%■purpose
■ 10.11%■amount
■  7.67%■housing
■  6.78%■years_at_residence
■  6.78%■other_credit
■  6.67%■job
■  3.78%■age
■  2.00%■phone
■  1.44%■dependents


Time: 0.0 secs

  [1] yes no  no  no  no  no  yes no  no  no  no  no  no  no  yes yes no  no
 [19] no  yes no  no  no  no  no  yes yes yes no  yes no  no  no  no  no  yes
 [37] no  no  yes no  no  no  no  no  no  no  no  no  no  no  yes no  no  no
 [55] yes no  no  no  no  yes no  no  no  no  no  no  no  no  no  no  yes no
 [73] no  yes no  no  no  no  no  no  yes no  no  yes no  no  no  no  yes yes
```

```
 [91] no  no  yes yes no  no  yes no  yes yes
Levels: no yes


   Cell Contents
|-------------------------|
|                       N |
|           N / Row Total |
|           N / Col Total |
|         N / Table Total |
|-------------------------|


Total Observations in Table:  100


                    | credit_pred
credit_test_labels |        no |       yes | Row Total |
-------------------|-----------|-----------|-----------|
                no |        55 |        13 |        68 |
                   |     0.809 |     0.191 |     0.680 |
                   |     0.733 |     0.520 |           |
                   |     0.550 |     0.130 |           |
-------------------|-----------|-----------|-----------|
               yes |        20 |        12 |        32 |
                   |     0.625 |     0.375 |     0.320 |
                   |     0.267 |     0.480 |           |
                   |     0.200 |     0.120 |           |
-------------------|-----------|-----------|-----------|
      Column Total |        75 |        25 |       100 |
                   |     0.750 |     0.250 |           |
-------------------|-----------|-----------|-----------|
```

Program 30 - Prediction of Stock Index Price Using Multiple Linear Regression Technique

```
Call:
lm(formula = Stock_Index_Price ~ Interest_Rate + Unemployment_Rate,
    data = economy_train)

Coefficients:
      (Intercept)      Interest_Rate  Unemployment_Rate
            948.0              376.3             -108.1


Call:
lm(formula = Stock_Index_Price ~ Interest_Rate + Unemployment_Rate,
    data = economy_train)

Residuals:
   Min     1Q Median     3Q    Max
-60.39 -27.94  -4.56  28.87  77.99

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)         948.04     584.69   1.621    0.126
Interest_Rate       376.31      70.19   5.362 7.92e-05 ***
Unemployment_Rate  -108.08      77.99  -1.386    0.186
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 41.06 on 15 degrees of freedom
Multiple R-squared:  0.9445,■Adjusted R-squared:  0.9371
```

```
F-statistic: 127.5 on 2 and 15 DF,  p-value: 3.843e-10

      19       20       21       22       23       24
947.3085 947.3085 968.9243 936.5006 936.5006 947.3085
```

Program 31 - Prediction of Heart Disease Using Multiple Linear Regression Technique

```
Call:
lm(formula = heart.disease ~ biking + smoking, data = heart_train)

Coefficients:
(Intercept)       biking      smoking
    15.0544      -0.2012       0.1766
Call:
lm(formula = heart.disease ~ biking + smoking, data = heart_train)

Residuals:
     Min       1Q   Median       3Q      Max
-2.13208 -0.45475  0.03819  0.44617  1.96346

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 15.054409   0.089550  168.11   <2e-16 ***
biking      -0.201211   0.001536 -130.99   <2e-16 ***
smoking      0.176556   0.003973   44.44   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6557 on 397 degrees of freedom
Multiple R-squared:  0.9794,■Adjusted R-squared:  0.9793
F-statistic:  9449 on 2 and 397 DF,  p-value: < 2.2e-16

       401        402        403        404        405        406        407
 5.4440913 16.4045370  8.1247793 11.2222658 17.2218484 16.1765847 17.6887189
       408        409        410        411        412        413        414
 8.3564160 16.1982005 12.7563907 15.0924516 18.9850910 14.1742892 14.3655897
       415        416        417        418        419        420        421
11.7241850 10.3909897  6.2896639  6.7918109 16.3524668 18.3637670 13.3703626
       422        423        424        425        426        427        428
12.6947968  4.0092881  7.0184952 13.4002144 16.0939930  4.0623532  3.1172051
       429        430        431        432        433        434        435
 9.5227354 14.7013267  7.3890152  4.1669639 16.0266136  9.2010838  9.4676583
       436        437        438        439        440        441        442
 6.2755312 13.3066359 12.5636695  3.9756125 12.1166711  1.7652189  8.5307497
       443        444        445        446        447        448        449
13.4043196 14.9591901 15.9157926 13.1615463 11.4825476 12.1298872  4.9826187
       450        451        452        453        454        455        456
13.7050107 14.5989392  0.9879520 15.5769839  4.1099892 12.2135721  6.6552168
       457        458        459        460        461        462        463
 5.2862327  5.1997790  7.2250413 10.1779322  2.9476886 13.3106128  9.5381695
       464        465        466        467        468        469        470
 0.3087257  8.8996927  7.0543066  4.0386046  7.1426975 17.4287724 17.2191801
       471        472        473        474        475        476        477
 4.3066830  8.7619174  4.7244257 12.2656077 11.9207261 11.2860542  6.7085675
       478        479        480        481        482        483        484
12.8651001 18.9542018 18.7626692  5.2048251  9.5644596 10.2275771  6.4960775
       485        486        487        488        489        490        491
10.6638974  8.1062162 14.0660828  5.5718601  6.7696940  3.8071385  5.8904251
       492        493        494        495        496        497        498
 3.0556301 12.0510042 10.3309421  9.7561330 14.5225832 10.1959780 13.0308220
```

Program 32 - Prediction of Medical Expenses Using Multiple Linear Regression Technique

```
Call:
lm(formula = expenses ~ age + sex + bmi + children + smoker +
    region, data = insurance_train)

Coefficients:
    (Intercept)              age          sexmale              bmi
       -12083.3            264.3           -288.5            339.9
       children        smokeryes  regionnorthwest  regionsoutheast
          410.2          23832.4           -439.9          -1291.3
regionsouthwest
        -1263.1


Call:
lm(formula = expenses ~ age + sex + bmi + children + smoker +
    region, data = insurance_train)

Residuals:
     Min       1Q   Median       3Q      Max
-11070.0  -2783.7   -926.3   1255.7  25270.9

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     -12083.26    1135.37 -10.643  < 2e-16 ***
age                264.26      13.40  19.721  < 2e-16 ***
sexmale           -288.53     377.41  -0.765  0.44474
bmi                339.91      32.57  10.436  < 2e-16 ***
children           410.24     156.89   2.615  0.00906 **
smokeryes        23832.38     475.70  50.099  < 2e-16 ***
regionnorthwest   -439.90     543.63  -0.809  0.41861
regionsoutheast  -1291.29     534.51  -2.416  0.01588 *
regionsouthwest  -1263.15     537.30  -2.351  0.01892 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5934 on 991 degrees of freedom
Multiple R-squared:  0.7569,■Adjusted R-squared:  0.7549
F-statistic: 385.7 on 8 and 991 DF,  p-value: < 2.2e-16

      1001       1002       1003       1004       1005       1006       1007
27583.4127 27654.6512  1476.9030  9110.7582  6981.4349  6457.5775  5793.6412
      1008       1009       1010       1011       1012       1013       1014
34262.0651  3547.7564 10944.6876  7087.9132 29195.2402 15715.3142 11261.9993
      1015       1016       1017       1018       1019       1020       1021
 6076.8905 11433.6305  1271.4416  5969.6590 15151.3352  4954.9571 12418.8576
      1022       1023       1024       1025       1026       1027       1028
28046.1886 35263.5807  -569.5267 14860.4921  3963.8578 25299.6560  -372.3569
      1029       1030       1031       1032       1033       1034       1035
11376.1261  4391.7943 31915.8844 36956.8261  5338.1408 23547.3657 16353.7672
      1036       1037       1038       1039       1040       1041       1042
 9972.1544 29403.8658 33976.6012  3258.4527  2297.3010 30084.2063   231.6483
      1043       1044       1045       1046       1047       1048       1049
27175.8495  2822.4706 14552.7358 31888.5500  7804.7607 34265.6128  2146.5830
      1050       1051       1052       1053       1054       1055       1056
33649.2764 12075.7742 13517.7324 11126.5799 33977.5689  1909.6591 11119.2972
      1057       1058       1059       1060       1061       1062       1063
 9161.3560  9322.7748  6390.6445  7550.5256  3559.5938 11306.9301 40154.6197
      1064       1065       1066       1067       1068       1069       1070
 7652.0091  4659.6924  6762.3706 12516.9864 12433.0878 11609.2476 12148.7269
      1071       1072       1073       1074       1075       1076       1077
```

32957.5060	14964.9141	3361.5809	12823.7289	10004.4702	5549.8525	10361.0507
1078	1079	1080	1081	1082	1083	1084
2025.4546	28336.8362	15657.2307	-682.7498	5516.1723	4421.5431	5938.6861
1085	1086	1087	1088	1089	1090	1091
15046.8523	29063.5583	12011.2723	12971.7583	16715.8293	8675.5107	34890.7344
1092	1093	1094	1095	1096	1097	1098
12590.4280	4788.7232	27456.1297	12962.3796	4970.4724	37929.9768	3629.3162
1099	1100	1101	1102	1103	1104	1105
12152.8573	5195.6415	27780.6742	11322.8506	7646.6893	13927.8087	6271.8972
1106	1107	1108	1109	1110	1111	1112
11514.1509	10597.2722	10422.6417	3843.2725	6376.3948	13575.9803	34490.9164
1113	1114	1115	1116	1117	1118	1119
33159.5800	5051.4838	2037.3103	12386.1812	10588.8749	28925.3827	31451.7929
1120	1121	1122	1123	1124	1125	1126
3416.4821	27237.0454	12287.5892	39074.6599	6473.2940	32768.3929	12650.0328
1127	1128	1129	1130	1131	1132	1133
11062.5399	8884.0913	6909.0842	-2003.1976	7096.3291	9922.3149	16382.4565
1134	1135	1136	1137	1138	1139	1140
7450.5062	6083.2556	11478.7044	7188.9148	1903.7312	5339.6805	27373.7635
1141	1142	1143	1144	1145	1146	1147
12560.4871	9799.9179	8817.0168	8456.1127	11377.5010	13300.9622	37201.9456
1148	1149	1150	1151	1152	1153	1154
3611.8955	8617.5434	9054.7966	2974.3227	15203.6994	34119.0750	9309.9525
1155	1156	1157	1158	1159	1160	1161
11295.8412	6184.6296	30445.3046	2254.3417	3599.7135	9080.1068	11004.2328
1162	1163	1164	1165	1166	1167	1168
10462.5263	7873.5577	2263.9136	8344.4886	6045.8978	15121.7584	3499.3112
1169	1170	1171	1172	1173	1174	1175
7606.5844	9257.2256	25927.3726	31745.1159	15669.4932	7996.3290	6586.7428
1176	1177	1178	1179	1180	1181	1182
1678.8013	33662.8916	6947.6619	5845.5984	28493.9823	13472.5547	3990.8115
1183	1184	1185	1186	1187	1188	1189
3559.2843	10311.2809	26630.0056	8348.5551	29645.8106	14969.0802	32111.9582
1190	1191	1192	1193	1194	1195	1196
2248.9775	7629.8327	6556.2839	14665.3053	12593.4064	3063.5668	3222.1104
1197	1198	1199	1200	1201	1202	1203
26534.1891	8575.4465	8146.9009	4435.7185	6052.9218	13888.4811	3916.4612
1204	1205	1206	1207	1208	1209	1210
12494.6711	25717.8965	2918.3587	14894.1324	31884.1610	30623.2815	14977.1522
1211	1212	1213	1214	1215	1216	1217
7606.5283	9054.3519	-317.3041	12534.4604	5645.9016	5688.8819	5432.1617
1218	1219	1220	1221	1222	1223	1224
7496.0763	34812.9333	9018.0123	3714.0238	6215.2443	8149.5259	24043.5657
1225	1226	1227	1228	1229	1230	1231
7010.4504	9291.3765	4206.0622	10894.0459	12838.4425	13256.1261	37714.5845
1232	1233	1234	1235	1236	1237	1238
23181.1470	11340.9592	9611.9180	8797.2091	4618.2904	11927.4417	12394.3543
1239	1240	1241	1242	1243	1244	1245
6354.1021	7962.4980	38939.3867	40465.3300	1754.4931	5279.0609	2422.7227
1246	1247	1248	1249	1250	1251	1252
4075.2552	8511.5819	6719.8452	4917.2637	31758.2179	27942.2613	-1883.8411
1253	1254	1255	1256	1257	1258	1259
24762.1108	8824.1972	5032.5038	10634.9834	14552.3079	11553.8089	15772.8863
1260	1261	1262	1263	1264	1265	1266
9537.2602	3347.9345	6785.1409	7404.7190	8590.2088	13020.1289	35158.0631
1267	1268	1269	1270	1271	1272	1273
11555.0189	28362.0485	3257.7181	8834.8836	5997.2663	5805.0534	8425.6782
1274	1275	1276	1277	1278	1279	1280
5381.0747	26237.9413	9483.6230	4063.6478	6040.2963	32348.6654	4009.9288
1281	1282	1283	1284	1285	1286	1287
10639.0094	33946.6147	23868.1909	1711.2604	39066.1206	8603.4639	1192.9958
1288	1289	1290	1291	1292	1293	1294

```
 5380.7652 29695.4781 10040.1877  5560.2097 27081.1581  -224.3777  9325.8538
      1295       1296       1297       1298       1299       1300       1301
11512.3947  -461.5473  1264.9707  3856.1364  6061.5017  1658.9374 32380.5501
      1302       1303       1304       1305       1306       1307       1308
39130.0985   740.3883 31010.0090 31743.3629  2389.9195 26839.6162 30676.1336
      1309       1310       1311       1312       1313       1314       1315
27357.6719  8965.1843  7642.0954  5271.2324 10342.1617 28122.1828 28508.2221
      1316       1317       1318       1319       1320       1321       1322
 2417.9074 -1323.3803  9152.9141 12015.9682  7548.0903  7170.2914 36918.4680
      1323       1324       1325       1326       1327       1328       1329
15919.5912 36099.2669  4606.0874 15146.8063 10188.3908 10431.7886  3049.4428
      1330       1331       1332       1333       1334       1335       1336
14047.4425 11257.9012  4084.4842 16819.6554 12158.9102  3523.2752  3907.7350
      1337       1338
  972.6616 37310.1300
```

Program 33 - Partition of Iris Data Set Using k-means Clustering Algorithm

```
K-means clustering with 3 clusters of sizes 62, 38, 50

Cluster means:
  sepal.length sepal.width petal.length petal.width
1     5.901613    2.748387     4.393548    1.433871
2     6.850000    3.073684     5.742105    2.071053
3     5.006000    3.428000     1.462000    0.246000

Clustering vector:
  [1] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
 [38] 3 3 3 3 3 3 3 3 3 3 3 3 3 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 [75] 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 2 2 2 2 1 2 2 2 2
[112] 2 2 1 1 2 2 2 2 1 2 1 2 1 2 2 1 1 2 2 2 2 2 1 2 2 2 1 2 2 2 1 2 2 2 1 2 2
[149] 2 1

Within cluster sum of squares by cluster:
[1] 39.82097 23.87947 15.15100
 (between_SS / total_SS =  88.4 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"

             1  2  3
  Setosa     0  0 50
  Versicolor 48  2  0
  Virginica  14 36  0
```

Program 34 - Partition of Breast Cancer Data Set Using k-means Clustering Algorithm

```
K-means clustering with 2 clusters of sizes 438, 131

Cluster means:
  radius_mean texture_mean perimeter_mean area_mean smoothness_mean
1    12.55630     18.57037       81.12347  496.0619       0.0948845
2    19.37992     21.69458      128.23130 1185.9298       0.1012946
  compactness_mean concavity_mean concave.points_mean symmetry_mean
1       0.09109982     0.06243776          0.03343254     0.1780580
2       0.14861298     0.17693947          0.10069878     0.1915397
  fractal_dimension_mean radius_se texture_se perimeter_se  area_se
1             0.06345402 0.3041909   1.215153     2.152881 23.78529
2             0.06060290 0.7428038   1.222538     5.250580 95.67817
  smoothness_se compactness_se concavity_se concave.points_se symmetry_se
1   0.007173263     0.02347469   0.02874551        0.01063632  0.02061358
```

```
2   0.006598687     0.03217669   0.04241977        0.01567398  0.02030397
  fractal_dimension_se radius_worst texture_worst perimeter_worst area_worst
1          0.003747503     14.04390      24.70954        91.93751   619.6479
2          0.003953389     23.70947      28.91267       158.49618  1753.0229
  smoothness_worst compactness_worst concavity_worst concave.points_worst
1        0.1299591         0.2233118       0.2192149           0.09132984
2        0.1404247         0.3577577       0.4493061           0.19243107
  symmetry_worst fractal_dimension_worst
1      0.2835537              0.08328194
2      0.3118817              0.08616550

Clustering vector:
  [1] 2 2 2 1 2 1 2 1 1 1 1 2 2 1 1 1 1 2 2 1 1 1 1 2 2 2 1 2 2 2 2 1 2 2 2 2 1
 [38] 1 1 1 1 1 2 1 1 2 1 1 1 1 1 1 1 2 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 2 1 2 1
 [75] 1 2 1 2 2 1 1 1 2 2 1 2 1 2 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 2 1 1
[112] 1 1 1 1 1 1 1 2 2 1 2 2 1 1 1 1 2 1 2 1 1 1 1 2 1 1 1 1 1 1 2 1 1 1 1 1 1
[149] 1 1 1 1 1 1 1 1 2 1 1 1 1 2 2 1 2 1 1 2 2 1 1 1 1 1 1 1 1 1 1 2 2 2 1 1
[186] 1 2 1 1 1 1 1 1 1 1 1 1 2 2 1 1 2 2 1 1 1 1 2 1 1 2 1 2 1 1 1 1 1 2 2 1 1
[223] 1 1 1 1 1 1 1 1 2 1 1 2 1 1 2 2 1 2 1 1 1 1 2 1 1 1 1 1 2 1 2 2 2 1 2 1 2
[260] 1 2 2 2 1 2 2 1 1 1 1 1 1 2 1 2 1 1 2 1 1 2 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1
[297] 1 1 1 1 2 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 2 1 2 1 1 1 1 1 1 1 1
[334] 1 1 2 1 2 1 2 1 1 1 2 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 2 2 1 2 2
[371] 1 1 2 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 2 2 1 1 1 1 1 1 2 1 1 1 1 1 1
[408] 1 2 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 1 1 1 1 1 1 1 2 1 1
[445] 2 1 2 1 1 2 1 2 1 1 1 1 1 1 1 1 2 2 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1
[482] 1 1 1 1 1 1 2 1 1 1 2 2 1 1 1 1 1 2 2 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 2 2
[519] 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 2 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
[556] 1 1 1 1 1 1 1 1 2 2 2 1 2 1

Within cluster sum of squares by cluster:
[1] 28559677 49383423
 (between_SS / total_SS =  69.6 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
[6] "betweenss"    "size"         "iter"         "ifault"

     1   2
  B 356   1
  M  82 130
```